Annie Prasanna Manoharan

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Education

University of Arkansas Aug 2024 – May 2026
Master of Science in Computer Engineering *Fayetteville, AR*
Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Privacy Enhancing Techniques

Sri Ramakrishna Engineering College Aug 2019 – May 2023
Bachelor of Engineering in Electronics and Communication *India*

Technical Skills

Data Engineering: Flink SQL, ETL/ELT, Apache Spark, Apache Kafka, Apache Airflow, Apache Flink, Databricks
Cloud and Tools: AWS Glue, Model monitoring, Feature engineering, Class imbalance handling, Cross-validation, PySpark, Experimentation and A/B testing, Amazon S3, Grafana, Docker, Git, Linux, Hyperparameter tuning, Vector embeddings
Machine Learning and AI: PyTorch, Scikit-learn, Computer vision, RAG, LangChain, Pandas, NumPy, LSTM, FAISS, Flask
Languages: SQL, Python

Projects

Transaction Anomaly Detection with Imbalanced Labels *Python, PySpark, Scikit-learn, Apache Kafka*

- Built a training pipeline in PySpark that aggregates raw transaction events into account-level and time-window features, with explicit label definition and leakage checks separating feature windows from label windows
- Trained and compared gradient-boosted and logistic classifiers under heavy class imbalance, using precision-recall evaluation and threshold analysis instead of accuracy
- Prototyped streaming scoring by publishing model outputs to Apache Kafka topics for downstream review queues

MiGa - Multi-Chicken Gait Assessment (AICV Lab) *YOLO11n-Pose, OC-SORT, LSTM, PyTorch*

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking, and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a detection split of 4,569 frames with 11,422 bounding boxes and a regression split of 55 expert-annotated sequences
- Implemented spatial normalization into local coordinate spaces to eliminate translational variance, achieving keypoint mAP 91.83% at 1,250 FPS, 0.93 MOTA, and 0.91 IDF1

Professional Experience

Data Engineering Intern Jul 2024 – Present
InsOps Inc. *Chicago, Illinois*

- Developed data pipelines in AWS using AWS Glue, Lambda, EMR, SQL, and Apache Airflow for orchestration, with S3 cloud storage, for scalable data ingestion, transformation, validation, and analytics
- Built data workflows integrating SQL Server, Azure SQL, AWS S3, and cloud databases, preparing structured datasets for downstream data science and analytics

Teaching Assistant Sep 2024 – May 2026
University of Arkansas, Dept. of Computer Science and Engineering *Fayetteville, AR*

- Supported 100+ students in Programming Foundations through laboratory instruction, debugging assistance, and mentorship in C++ programming fundamentals

Research Assistant Sep 2024 – Jan 2026
AICV Lab, University of Arkansas *Fayetteville, AR*

- Built reusable preprocessing workflows to segment, organize, and transform raw image/video data into structured inputs for downstream processing and evaluation
- Evaluated results using mAP, MOTA, IDF1, PSNR, SSIM, and FID

Project Engineer Jun 2023 – Jun 2024
Soliton Technologies *India*

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to streamline application usage and improve test automation
- Developed customized software solutions for Dräger as the sole developer, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements